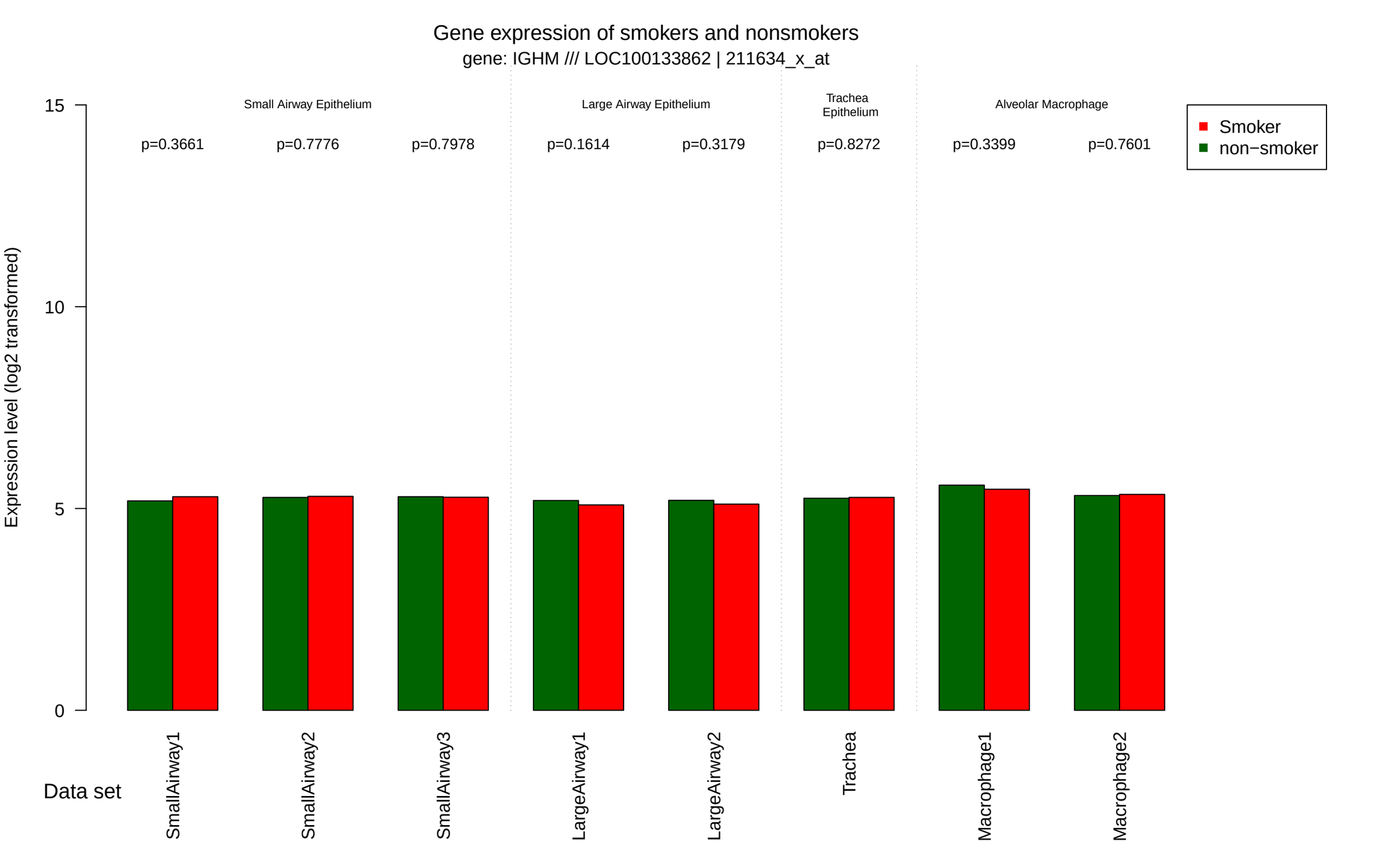
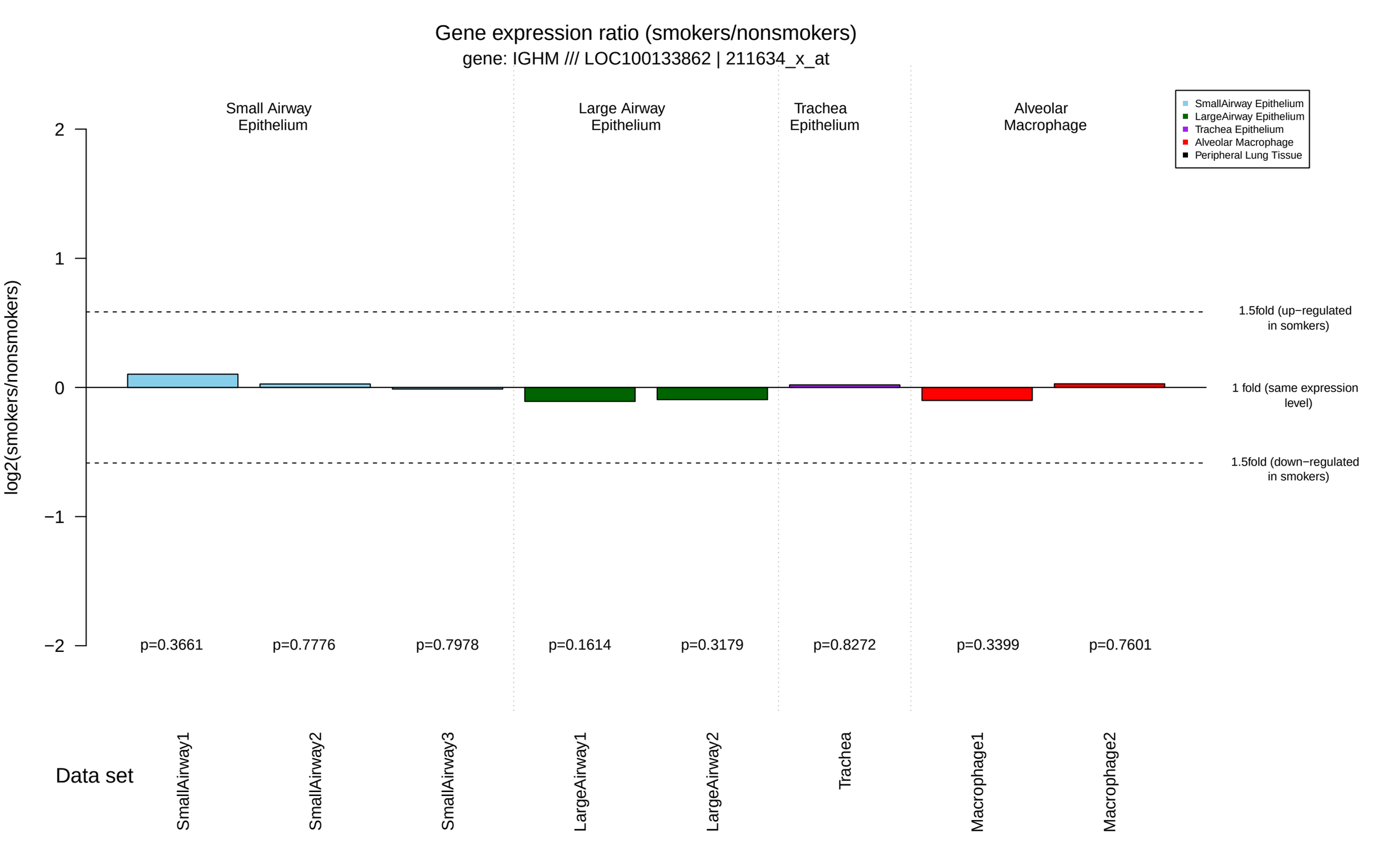
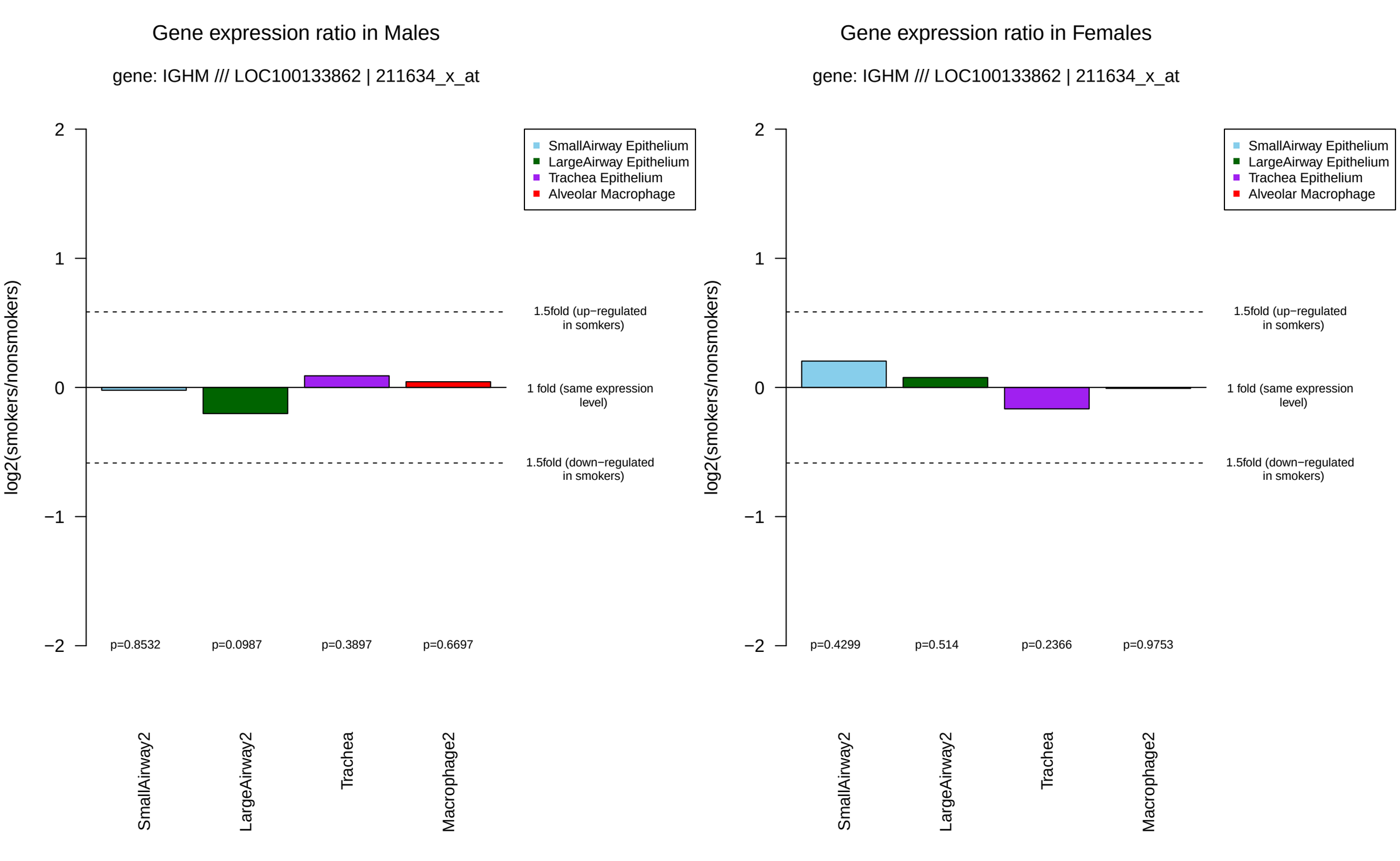
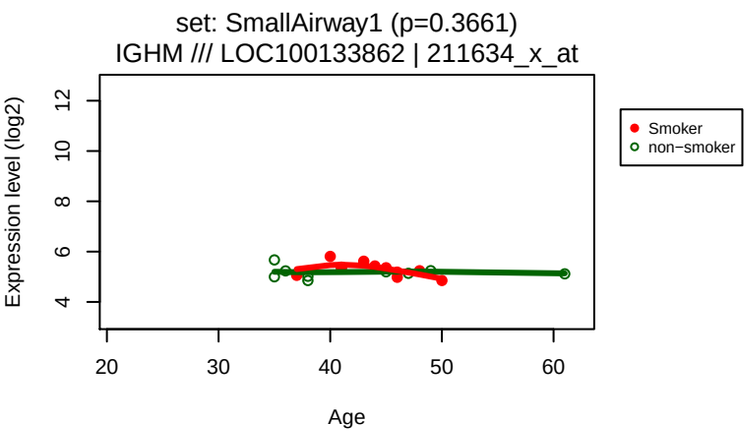
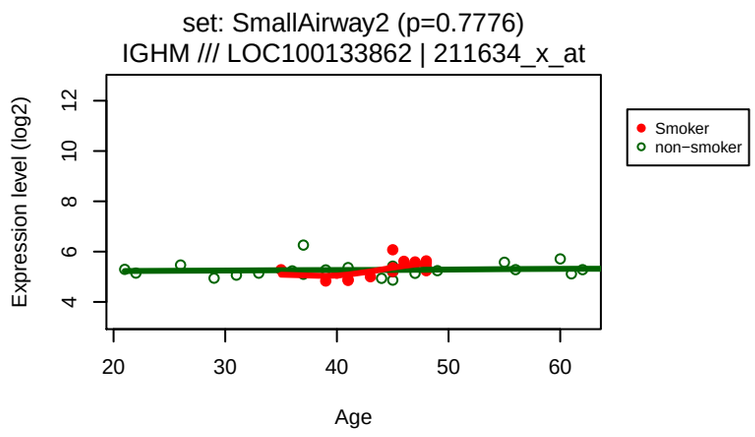
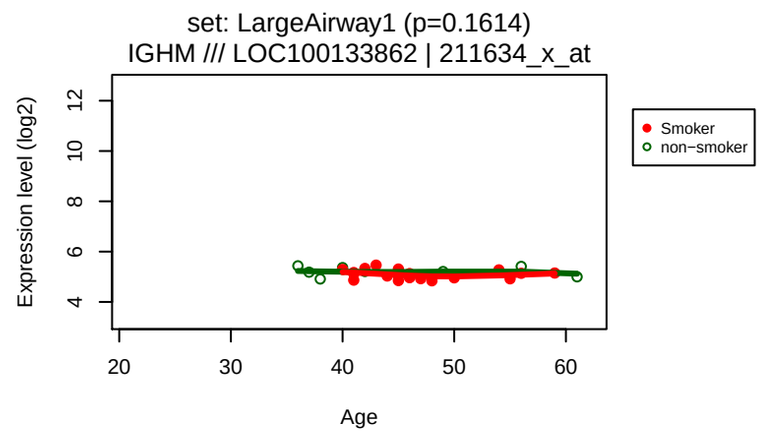

Gene expression of smokers and nonsmokers
gene: IGHM /// LOC100133862 | 211634_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3661
p=0.7776
p=0.7978
p=0.1614
p=0.3179
p=0.8272
p=0.3399
p=0.7601
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

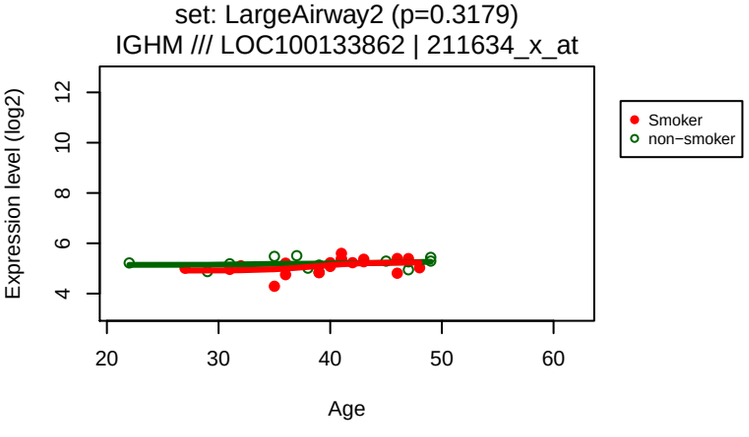

Gene expression ratio (smokers/nonsmokers)
gene: IGHM /// LOC100133862 | 211634_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3661
p=0.7776
p=0.7978
p=0.1614
p=0.3179
p=0.8272
p=0.3399
p=0.7601
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

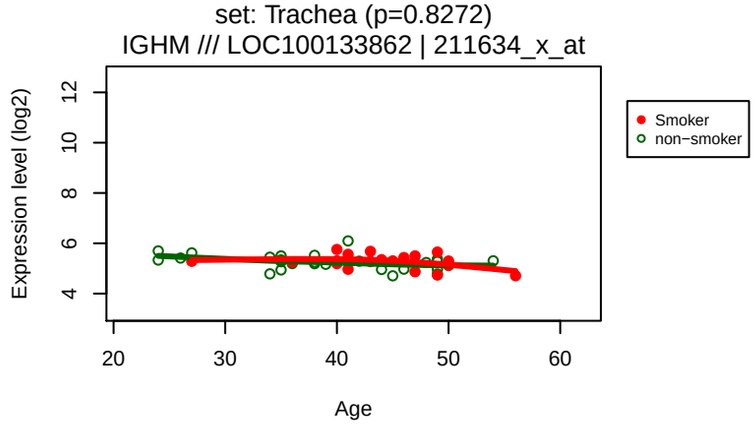

Gene expression ratio in Males
gene: IGHM /// LOC100133862 | 211634_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8532
p=0.0987
p=0.3897
p=0.6697
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHM /// LOC100133862 | 211634_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4299
p=0.514
p=0.2366
p=0.9753
SmallAirway2
LargeAirway2
Trachea
Macrophage2

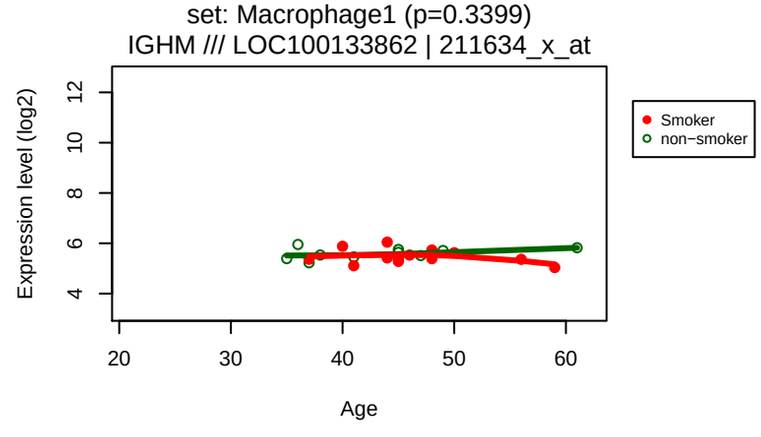

set: SmallAirway1 (p=0.3661)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker

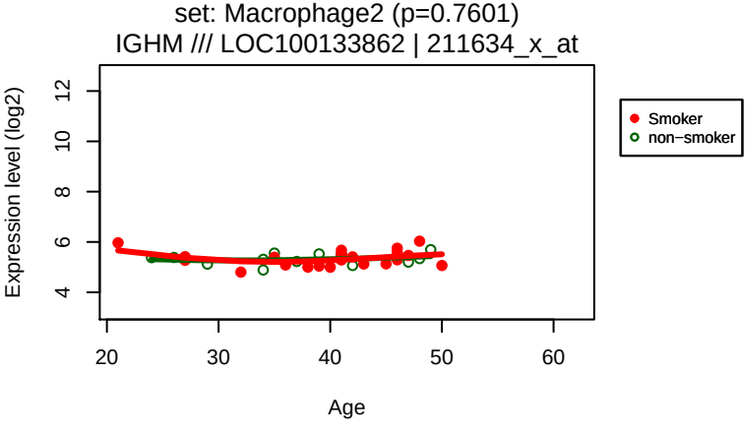

set: SmallAirway2 (p=0.7776)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker

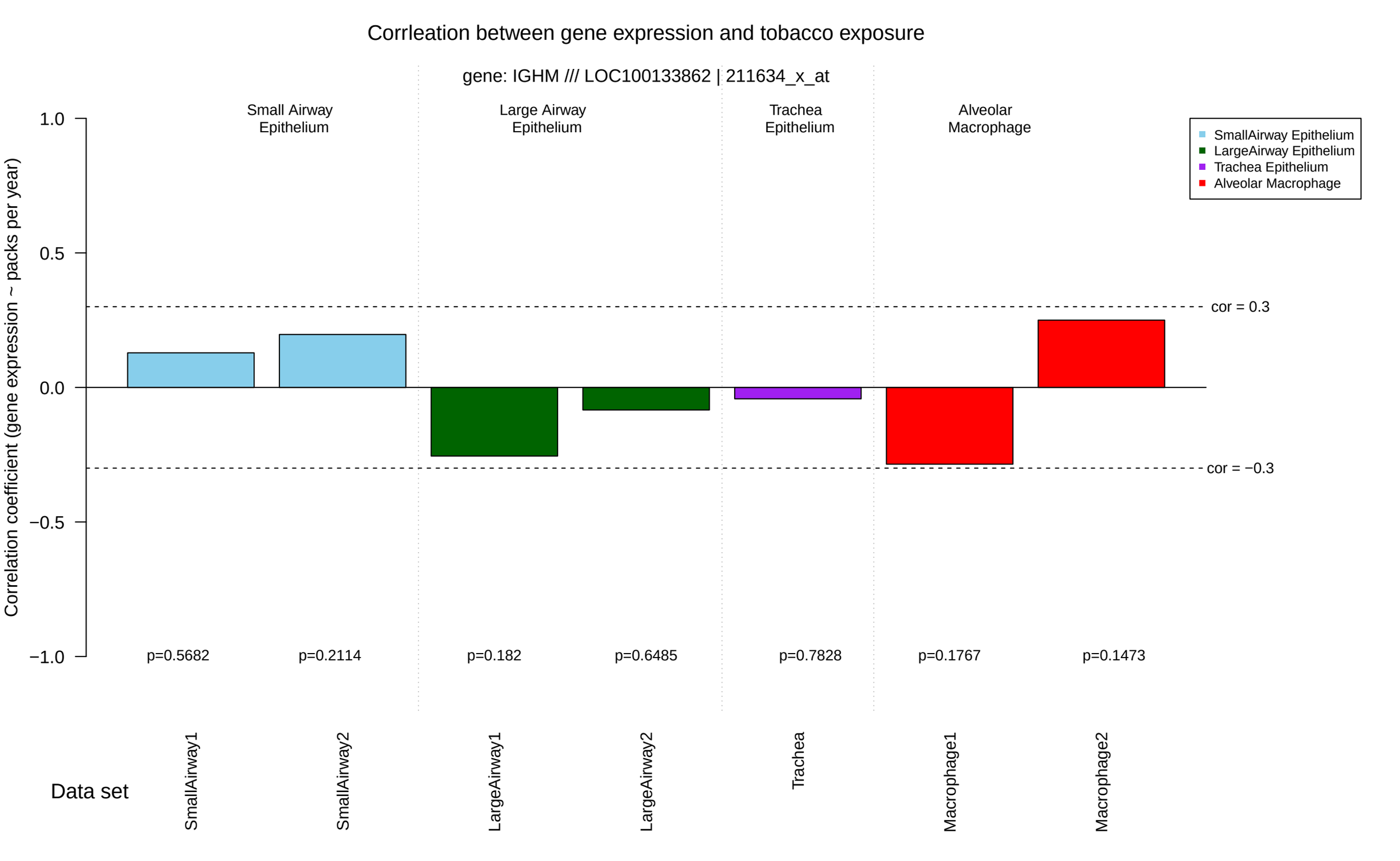

set: LargeAirway1 (p=0.1614)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3179)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8272)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3399)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7601)
IGHM /// LOC100133862 | 211634_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHM /// LOC100133862 | 211634_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5682
p=0.2114
p=0.182
p=0.6485
p=0.7828
p=0.1767
p=0.1473
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2